

SELECTED WORK

Willon Smith

AI Engineer | Technical Delivery Lead

Production AI systems, full stack platforms and automation delivered end to end. I pick work up while the business requirement is still unclear and stay with it through architecture, build, deployment and production support.

Client names are withheld under NDA. Every figure in this document was measured in production or in a verified test run.

7

UK client businesses delivered for

260+

backend tests behind an 85% CI coverage gate

99.8%

completion on an 80,000 record AWS ingestion

0

wrong name errors across a 180 message concurrency regression

RECENT WORK

AI Engineer and Technical Delivery Lead, UK AI consultancy · April 2026 to October 2026

Business Operating Platform

FULL STACK PLATFORM · SOLE ENGINEER · LIVE IN PRODUCTION

A whole business process platform for a UK construction client, replacing spreadsheet and board based tracking. Built from an unclear brief with no requirements document, delivered in vertical slices, each proving value in production before the next began.

- Django
- DRF
- React
- TypeScript
- PostgreSQL
- DuckDB
- Docker
- GitHub Actions

- Process engine with approval and evidence gates, including value thresholds that route higher value runs to director approval
- Four layer, account scoped role based access control, adversarially audited for cross tenant isolation
- Dynamic form builder reproducing the client's real contract paperwork field for field
- Analytics warehouse feeding an executive view of spend, exposure and compliance

IMPACT

Live in production with 260+ backend tests behind a CI enforced 85% coverage gate. On its first sync of real data the analytics view surfaced £1,726 a week of live plant hire spend and around £600 a week of forgotten off hire charges.

Permission Aware Multi Channel AI Assistant

AI AGENTS · CLAUDE TOOL CALLING · THREE CHANNELS

An assistant that answers operational questions from company data and raises requests, built so it can never see or do more than the person asking it. One agent core serves every channel, so each new channel inherits the same security model.

- Claude / Anthropic API
- Python
- DuckDB
- Telegram Bot API
- WhatsApp
- Django

- Guarded read only SQL layer with view allowlists and per tenant query rewriting
- Write actions require explicit user confirmation before executing
- Agent permissions inherit platform identity, so the model cannot exceed the initiating user's authority
- Charts and answers returned in channel, including image to record capture using vision

IMPACT

Live across in app, Telegram and WhatsApp. An adversarial permission parity audit found two data leak paths before release. Both were closed and are covered by regression tests.

Asset Analytics Platform and AI Query Agent

DATA PLATFORM · NATURAL LANGUAGE ANALYTICS · RAPID DELIVERY

A modelled analytics warehouse and portal for a brand activation client working with global drinks brands, with a natural language query agent over the top. Rebuilt for a second brand on the same chassis inside a week to meet a tender deadline.

- dbt
- PostgreSQL
- Vercel
- Claude
- SQL
- Python

- Live portal delivered within ten days of project confirmation
- Query agent fenced by a least privilege read only database role, with tests proving the fence holds
- Release gated by a fifteen case golden evaluation suite that blocks on failure
- Fully isolated schemas per brand with a no cross access role model

IMPACT

A 44 check reconciliation, one per on screen figure, verified the entire interface against live SQL before the tender went in.

RECENT WORK CONTINUED

Deterministic Email Campaign Engine

PRODUCTION AUTOMATION · CORRECTNESS CRITICAL SEND PATH

A client wanted AI writing customer facing emails in a live send path. The brief demanded provable correctness, so I designed a deterministic engine and kept the model out of the path where being wrong cannot be undone. Only the client's own words are ever sent.

n8n Microsoft Graph Airtable JavaScript OAuth 2.0

- An 89 node legacy workflow re-architected into a four workflow system: campaign portal, follow up engine, reply and bounce handling, live monitoring
- Code enforced guardrails: hard follow up cap, per run circuit breaker, per recipient cooldown, duplicate stage guard
- An assertion that throws before sending if the greeting does not match the recipient
- Bounce and out of office detection so only genuine delivery failures stop a sequence

IMPACT

A 180 message concurrency regression across four simultaneous campaigns passed with zero wrong name errors and no duplicates before go live.

Large Scale Data Ingestion on AWS

DATA ENGINEERING · VECTOR SEARCH · INFRASTRUCTURE LIFECYCLE

Monthly refresh of a five million record image matching database for a consumer application, executed end to end from cost estimate through to clean infrastructure teardown.

AWS ECS RDS pgvector Terraform Celery Python S3

- Filtered a 10.3 GB, 19.2 million record dataset to a candidate set, diffed against the live database
- Ran an 80,000 record ingestion on ECS workers against RDS with pgvector HNSW indexing
- Wrote a streaming parser sustaining around 8,000 records per second, memory safe on a constrained host
- Infrastructure raised and torn down per run via targeted Terraform applies to avoid destructive drift

IMPACT

99.8% completion with before and after evidence for sign off. I root caused a mid run throughput collapse to provider throttling and documented a repeatable runbook, so the process no longer depends on its original author.

Production Resilience and Monitoring

PRODUCTION SUPPORT · OBSERVABILITY · INCIDENT RESPONSE

Inherited a live automation estate that had suffered an eighteen hour silent outage, with five of seven workflows running with no error alerting at all. Rebuilt the operational layer around it.

n8n Microsoft Graph Slack Python Grafana Loki

- Error alerting and heartbeat stuck detection built from scratch, running at zero incremental cost
- Around fifty integration nodes retry hardened in a single audit pass
- Fragile polling triggers replaced with scheduled fetch and duplicate detection, structurally removing the stall class
- Lost records recovered by API driven execution retries

IMPACT

Recurring stalls eliminated. Monitoring became a standard requirement on every deployment that followed.

RECENT WORK CONTINUED

Document RAG Application Repair

PRODUCTION FIX · DOCUMENT RAG · FASTAPI

A FastAPI and MongoDB Atlas application that a client uses to answer tender questionnaires from its own document library. Others built it; I took on its production faults after a 336 page manual would not upload and other uploads hung for five minutes or more.

- FastAPI
- MongoDB Atlas
- Docling
- Embeddings
- Docker

- Traced the hang to parsing and one embedding call per chunk running inside the request, with the blocking parser stalling every other request during an upload
- Uploads now return at once: parsing runs in a background task on a bounded worker thread, embeddings go in batches of 64, and each document shows its status
- Six pull requests in five days, adding scanned document and legacy .doc support and a migration that declares the vector index filters
- A review pass over the areas the fix had not touched found six more bugs, all fixed

IMPACT

Uploads that hung for five minutes or more returned in under three seconds, verified in production on the exact files users had reported.

Company Website Redesign

SEO VERIFICATION · MARKETING SITE · IN HOUSE

The consultancy's own website, whose search work had been bought from an outside supplier. I checked what the supplier had actually done, then designed and built a redesign that kept every page the site ranked for.

- HTML
- CSS
- JavaScript
- Python
- Structured data

- Checked each of the supplier's 37 claimed actions on the live HTML and on the page rendered at desktop and phone width, with the evidence beside each status
- Found five further problems the supplier's review had missed, including a sitemap that listed the 404 page
- Built the 12 page redesign from one template set with a Python builder that refuses to write a page with a placeholder left in it
- Captured a baseline from the live site and kept every ranking page at its exact URL, with a byte for byte title match and structured data on every page

IMPACT

Of the 37 claimed actions, 20 were done, 5 were done incorrectly and 12 were not done. The redesign's design carried into the site that went live in August 2026.

EARLIER WORK

Workflow automation and application development for SME clients · 2021 to 2026

Enterprise Payment Cycle System

FINANCIAL AUTOMATION · MULTI BRANCH DEPLOYMENT

A complete payment request, document filing and proof of payment reconciliation system deployed across five or more business branches, replacing a manual forms and scripts process.

n8n Google Sheets API Drive API Webhooks Gmail

- Auto generated document numbers tracked across every branch
- File categorisation that detects the document type from filename patterns
- Dedicated webhook endpoints and mail accounts per branch
- Real time status tracking with links back to every stored document

IMPACT

Manual document tracking eliminated across all branches, with an audit trail on every transaction that the banking team cross references before approving payment.

Automated Proof of Payment Processing

EMAIL AUTOMATION · DOCUMENT PROCESSING · FRAUD CHECKS

An email monitoring system that watches branch inboxes for incoming bank proof of payment messages, extracts the payment data, matches it against existing requests and files everything automatically.

n8n IMAP Google Drive Google Sheets Pattern matching

- Security validation layer applying fraud prevention checks to every inbound message
- Duplicate detection preventing double processing
- Document number matching linking each payment proof to its original request
- Client folders created on the fly where none existed

IMPACT

Manual filing removed entirely and reconciliation time reduced from hours to minutes, with no missed proofs of payment.

AI Care Assistant

AI CHATBOT · HEALTHCARE OPERATIONS

An AI assistant for a care facility handling resident communication, vital sign recording, care plan queries and shift summaries, synchronising data across several platforms.

Claude n8n Telegram API Google Sheets

- Natural language interface over a chat channel staff already used
- Vital sign recording and tracking against individual care plans
- Automated shift summary generation and distribution
- Multi platform data synchronisation keeping records aligned

IMPACT

Reduced administrative load on care staff and improved handover between shifts.

EARLIER WORK CONTINUED

Paycycle and Auditor Application

FULL STACK APPLICATION · MULTI LEVEL APPROVALS

A paycycle management and auditing application covering the full payment lifecycle from request through approval to audit trail, deployed across multiple branches with role based access.

Python Supabase n8n Google APIs

- Multi level approval workflow with automated document generation and numbering
- Branch specific configuration on a shared deployment
- Complete audit history on every transaction
- Reporting dashboard and stage by stage email notification

Workshop Job Card and Costing System

BUSINESS PROCESS AUTOMATION · INDUSTRIAL SERVICES

Job card and costing system for an industrial services company, managing container unpacking, equipment processing, costing calculations and invoice generation across multiple sites.

n8n Google Sheets PDF generation Custom forms

- Work order creation, tracking and completion
- Automated costing and pricing calculations
- Tax invoice generation directly from job card data

CRM and Business Process Suite

CRM CONFIGURATION · REPORTING AUTOMATION

A CRM and automation suite integrating Monday.com with Google Workspace for project management, reporting and client relationship tracking.

Monday.com Apps Script AppSheet n8n

- CRM customisation, pipelines and automations for multiple clients
- Automated reporting from time tracking data
- Leave management with approval workflow, and a custom Kanban application

Enterprise Document Management

DOCUMENT MANAGEMENT · WORKFLOW DIGITISATION

Implemented and administered an enterprise document management system for a real estate business, converting paper based processes into routed digital workflows.

DocuWare Windows Server Merge forms

- Workflow, merge form and web form design
- User role structures and access control
- Technical documentation and user training for adoption

TECHNICAL SUMMARY

AI and LLMs	Claude and the Anthropic API, Claude Code daily in production, AI agents and tool calling, RAG and retrieval pipelines, embeddings and vector search, prompt caching, multi tier model routing, golden evaluation suites
Languages	Python, TypeScript, JavaScript, SQL, HTML and CSS, Google Apps Script
Frameworks	Django and Django REST Framework, FastAPI, Flask, React, Node.js
Data	PostgreSQL, pgvector, DuckDB, dbt, MongoDB Atlas, Supabase, warehouse modelling, reconciliation frameworks
Cloud and infra	AWS (ECS, RDS, S3), Terraform, Docker, GitHub Actions, Vercel, Neon, Render, Hetzner
Security and ops	Linux server hardening, NGINX, TLS, UFW, fail2ban, Tailscale, systemd, OAuth 2.0, role based access control, database role fencing, least privilege, adversarial permission testing
Observability	Grafana, Loki, heartbeat monitoring, alerting, incident response, spend controls and kill switches
Automation	n8n at depth, webhooks, IMAP triggers, scheduled workflows, API integration, circuit breakers and idempotency
Integrations	Microsoft Graph and Outlook, Business Central OData, Monday.com, Airtable, Linear, Telegram Bot API, Twilio and WhatsApp, Calendly, Google Workspace, AppSheet

A note on this document. Client identities are withheld under NDA and no client system, screenshot or live link is reproduced here. Every metric quoted was measured in production or in a verified test run. Where work was led by a colleague and I contributed, this document does not claim it. I am happy to go deeper on the engineering behind any of these at interview.

Willon Smith

AI Engineer and Technical Delivery Lead
+27 79 706 1500 · willonsmith7@gmail.com
linkedin.com/in/willon-smith-583361164 · github.com/willon-smith
willon-smith.github.io · Bloemfontein, South Africa
Open to remote permanent roles · SAST (GMT+2), strong UK and European hours overlap